

Software Requirements Specification (SRS) Template

Project: Charity Event Fundraiser Tracking System

Version: 1.0

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Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the requirements for the Charity Event Fundraising Tracker system. The purpose of this system is to provide event organizers with a centralized platform to create, manage, and monitor fundraising events while allowing donors to make pledges seamlessly. It ensures real-time tracking of fundraising goals, transparency for donors, and operational efficiency for organizers. This document serves as a reference for developers, testers, and stakeholders to align expectations and verify successful system implementation.

1.2 Scope

The Charity Event Fundraising Tracker supports the complete fundraising lifecycle, from event creation to pledge collection and goal achievement. Its main features include:

- Event creation with fundraising goals and deadlines.
- Secure donor pledge submissions (with or without login).
- Real-time progress tracking with graphical visualization.
- Role-based dashboards for organizers, administrators, and donors.
- Reporting and analytics for donor contributions and fundraising outcomes.
- Exportable reports for accountability and transparency.

The system focuses on accessibility (web and mobile-friendly), scalability to support large fundraising campaigns, and secure handling of donor data. It excludes direct financial transaction processing (payments will be linked to third-party gateways).

1.3 Audience

- Developers: To design and implement system modules based on requirements.
- Event Organizers: To create and manage fundraising events and monitor progress.
- Donors: To pledge contributions and track fundraising results transparently.
- System Administrators: To oversee user accounts, ensure compliance, and manage data integrity.
- Stakeholders (NGOs, Charitable Organizations, Sponsors): To evaluate fundraising effectiveness and ensure accountability.

1.4 Definitions

- Pledge: A donor's commitment to contribute a specific amount towards an event.
- Target Goal: The fundraising objective (amount) set for an event.
- Progress Bar: A visual representation of the percentage of funds raised compared to the target.
- Event Summary Report: A compiled document showing donors, pledges, and fundraising status.
- Dashboard: Role-specific interface displaying real-time event statistics and controls.

2. Overall description

2.1 Product Perspective

The Charity Event Fundraising Tracker is a web-based system that connects donors and event organizers through a user-friendly interface. It interacts with:

- A database layer for storing donor pledges, event data, and reports.
- Charting and visualization tools for real-time progress tracking.
- Third-party payment gateways (optional integration) for processing donations.

The system is designed as a standalone application but can be extended to integrate with external NGO or charity platforms via REST APIs.

2.2 Major Product Functions

- Event Management: Create, edit, and close charity fundraising events.
- Donor Pledge Management: Collect, validate, and securely store donor pledges.
- Progress Tracking: Aggregate pledges, display fundraising progress via dashboards and charts.
- User Access Control: Provide role-based access (Organizers, Donors, Admins).
- Reporting & Analytics: Generate event summaries, donor leaderboards, and export reports.
- Notifications: Send automated updates on event milestones and deadlines.

2.3 User Roles and Characteristics

- Donors: Occasional users with minimal technical skills, expect a quick and simple pledge submission process.
- Event Organizers: Frequent users, require comprehensive dashboards, goal monitoring, and reporting tools.
- Administrators: Technical staff, responsible for user management, system monitoring, and compliance checks.
- Sponsors/Partners (Optional): May need read-only access to performance and transparency reports.

2.4 Operating Environment

- Web-based application: Accessible on modern browsers (Chrome, Edge, Firefox, Safari).
- Mobile support: Responsive design for smartphones and tablets.

- Server environment: Hosted on a cloud platform with scalable backend (e.g., AWS/Azure).
- Database: Relational database (e.g., MySQL/PostgreSQL) for structured event and donor data.
- Connectivity: Secure HTTPS connections with TLS 1.2+ encryption.

2.5 Constraints

- Must comply with data privacy regulations (e.g., GDPR for donor data protection).
- Must ensure role-based security for donor and organizer accounts.
- Should support scalability for large events (up to 10,000+ donors).
- Relies on third-party payment providers for actual transaction handling; does not store credit card details.
- Budget and timeline constraints may limit advanced analytics features in the initial release.

3. External Interface Requirements

3.1 User Interfaces

The system shall provide two primary interfaces:

Donor UI

- Web-based, mobile-responsive pledge form.
- Simple fields: Name, Email/Phone (optional), Pledge Amount, Message (optional).
- Real-time confirmation message upon pledge submission.
- Accessibility: Screen-reader compatibility (WCAG 2.1 AA), high-contrast mode, and keyboard navigation support.

Organizer Dashboard

- Secure login screen (username + password).
- Event management panel to:
 - Create/Edit/Delete events with details (name, description, target goal, dates).
 - Monitor fundraising progress through graphical charts and progress bars.
 - View donor list and export reports (CSV/PDF).
- Responsive design for desktop, tablet, and mobile devices.

Admin UI

- User management interface (add/remove organizers).
- View and audit all events and pledge records.
- Access to system logs and monitoring metrics.

3.2 Hardware Interfaces

Since this is a web-based application, specialized hardware is not required beyond standard computing devices. Interfaces include:

- Client devices: Desktops, laptops, tablets, smartphones (minimum screen size 5”).
- Server hardware: Hosting machine/cloud server capable of supporting database + web server.
- Optional hardware for on-site events:
 - Kiosk/touchscreen for live donor pledges.
 - QR-code scanners for quick donor access (future integration).

3.3 Software Interfaces

The system shall interact with:

- Database System (SQL/MongoDB): For storing event details, donor pledges, and reports.
- Authentication API: For secure login (e.g., OAuth2 for organizers and admins).
- Visualization Library: Chart.js / Recharts for progress bars and charts.
- Reporting Module: Exports data into CSV/PDF formats for organizers.
- Optional Payment Gateway API (future scope): Integration with Razorpay/PayPal/Stripe to convert pledges into actual donations.

3.4 Communications

- Protocol: All client-server communication via HTTPS (TLS 1.2/1.3).
- API Requests: RESTful JSON-based API for pledge submissions, event updates, and dashboard queries.
- Performance Handling: Asynchronous request handling to ensure <2s response for pledge submissions.
- Scalability: System should handle high loads (10,000 concurrent pledges).
- Notification Support: Optional email confirmation to donors via SMTP or third-party mail APIs (SendGrid/Mailgun).
- Real-time Updates (Optional): WebSockets or Server-Sent Events (SSE) for live event progress tracking.

4. System features

4.1 Event Management

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CFT-F-001	The system shall allow organizers to create fundraising events with details (event name, description, target goal, and duration).	Functional	High	Organizer	AC-CFT-F-001: Event created with all mandatory details visible in dashboard. Test: TC-Event-01	Requires authenticated organizer login.
CFT-F-002	The system shall allow organizers to edit or delete events before closing.	Functional	High	PCI-DSS	AC-CFT-F-002: Edited event updates correctly; deleted event removed from dashboard. Test: TC-Event-02	Deletion restricted if pledges already exist.
CFT-F-003	The system shall allow organizers to close events once fundraising is complete or end date is reached.	Functional	High	Security	AC-CFT-F-003: Closed events no longer accept pledges; status updated to "Closed". Test: TC-Event-03	Dependent on pledge module; closed status should lock submission form.

4.2 Donor Pledges

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CTF-F-004	The system shall allow donors to submit a pledge amount via a secure online form.	Functional	High	Donors / Organizers	AC-CFT-F-010: Donor fills in pledge form then the system validates and stores pledge. Test: TC-Pledge-01	Requires database and donor form UI

CTF-F-005	The system shall validate pledge data (amount > 0, donor details required).	Functional	High	System / Donors	AC-CFT-F-011: Invalid pledges (negative/blank data) rejected with error message. Test: TC-Pledge-02	Depends on input validation rules
CTF-F-006	The system shall update the fundraising event's progress in real time after a pledge.	Functional	High	Organizers	AC-CFT-F-012: After pledge, progress bar and totals update instantly. Test: TC-Pledge-03	Requires event-progress dashboard

4.3 Progress Tracking

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CFT-F-007	The system shall automatically aggregate the total number of pledges for each fundraising event.	Functional	High	Organizer, Donor	AC-CFT-F-007: Dashboard shows accurate total amount matching pledge records. Test: TC-Track-01	Depends on donor pledge module for accurate data.
CFT-F-008	The system shall display a progress bar showing percentage completion against the event target.	Functional	High	Organizer, Donor	AC-CFT-F-008: Progress bar updates in real time as new pledges are added. Test: TC-Track-02	Requires event target to be set in Event Management module.
CFT-F-009	The system shall generate visual reports (e.g., pie charts, bar graphs) to illustrate donation distribution	Functional	Medium	Organizer	AC-CFT-F-009: Reports generated correctly and downloadable in CSV/PDF. Test: TC-Track-03	Depending on Charting/reporting libraries.

	and trends.				TC-Track-03	
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4.4 User Role & Access

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CFT-F-010	The system shall allow the organizer to login.	Functional	High	Organizer / Security Team	AC-CFT-F-010: Valid credentials grant access to organizer dashboard. Test: TC-Org-Login-010	Requires authentication service
CFT-F-011	The system shall allow users to change	Functional	High	All/Security Team	AC-CFT-F-011: Password requirements met Test: TC-Password-011	Requires email/SMS OTP service
CFT-F-012	The system shall allow the admin to immediately revoke user access in case of policy violation or security risk	Functional	High	Admin / Security	AC-CFT-F-012: Revoked user cannot access any system functions. Test: TC-Revoked-012	Requires RBAC and admin override

4.5 Reports and Analytics

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CFT-F-013	Export donor pledge lists	Functional	High	Organizers	AC-CFT-F-013: After user exports the pledge list, the system should generate files with accurate data. Test: TC-LIST-013	Needs pledge module.
CFT-F-014	Generate Top Donors report	Functional	High	Organizers	AC-CFT-F-014: System should create a report of donors sorted by amount. Test: TC-Report-014	Depends on pledge data.
CFT-F-015	Event summary report	Functional Requirement	High	Organizers	AC-CFT-F-015: It will show total pledges collected and percentage of target achieved. Test: TC-SUM-015	Depends on pledge data.

5. Non-functional requirements

Req ID	Requirement		Category	Priority	Acceptance criteria / Measurement
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CFT-NF-001	The system shall provide 99.5% availability excluding scheduled maintenance windows.	Non Functional	Reliability	High	AC-CFT-NF-001: Uptime monitoring reports show $\geq 99.5\%$ availability per month. Test: Test: TC-Avail-01
CFT-NF-002	The system shall ensure pledge submission response time ≤ 2 seconds under normal load.	Non Functional	Performance	High	AC-CFT-NF-002 95th percentile of pledge submissions completes in ≤ 2 seconds during load testing. Test: TC-Perf-01
CFT-NF-003	The system shall be able to handle concurrent access and transactions from up to 10,000 donors without performance degradation.	Non Functional	Scalability / Performance	High	AC-CFT-NF-003: System supports 10,000 concurrent donors with average response time $\leq 3s$. Test: TC-Person-01
CFT-NF-004	Daily automated backup	Non-Functional	Admins, Organizers	High	AC-CFT-NF-004 System should automatically create backup everyday. Test:TC-BKP-01
CFT-NF-005	The system shall provide a responsive UI that adapts to desktops, tablets, and smartphones.	Non-Functional	Usability / Accessibility	Medium	AC-CFT-NF-005: UI verified on at least 3 screen sizes (desktop, tablet, mobile) with no functional loss. Test: TC-UX-01

5.1. Security

5.1.1 Security Objectives

1. Confidentiality of Sensitive Data

Ensure that donor, organizer, and admin information (personal and financial details) is protected from unauthorized access or disclosure.

2. Integrity of Transactions and Records

Safeguard the accuracy and consistency of donation, pledge, and fundraising event data against unauthorized modifications.

3. Availability of the System

Guarantee that the fundraising platform remains accessible and operational for donors, organizers, and admins, even under high traffic or attempted attacks.

5.1.2 Security Requirements

Req ID	Requirement	Type	Priority	Acceptance criteria / Test case ref
CFT-SR-001	The system shall encrypt sensitive donor and transaction data at rest and in transit.	Security	High	AC-SR-001: DB & API encryption verified; Test: TC-SR-Enc-01
CFT-SR-002	The system shall enforce multi-factor authentication (MFA) for organizers and admins.	Security	Medium	AC-SR-002: MFA required for login; Test: TC-SR-MFA-0
CFT-SR-003	The system shall log and audit all critical security events (login, logout, role changes, donations).	Security	High	AC-SR-003: Audit log entries generated; Test: TC-SR-Log-01
CFT-SR-004	The system shall lock user accounts after 5 failed login attempts, requiring admin reset.	Security	High	AC-SR-004: Account lockout triggered; Test: TC-SR-Lock-01
CFT-SR-005	TLS 1.2+ shall be mandatory for all network connections.	Security	High	AC-SR-005: All requests use HTTPS; Test: TC-SR-Conn-01

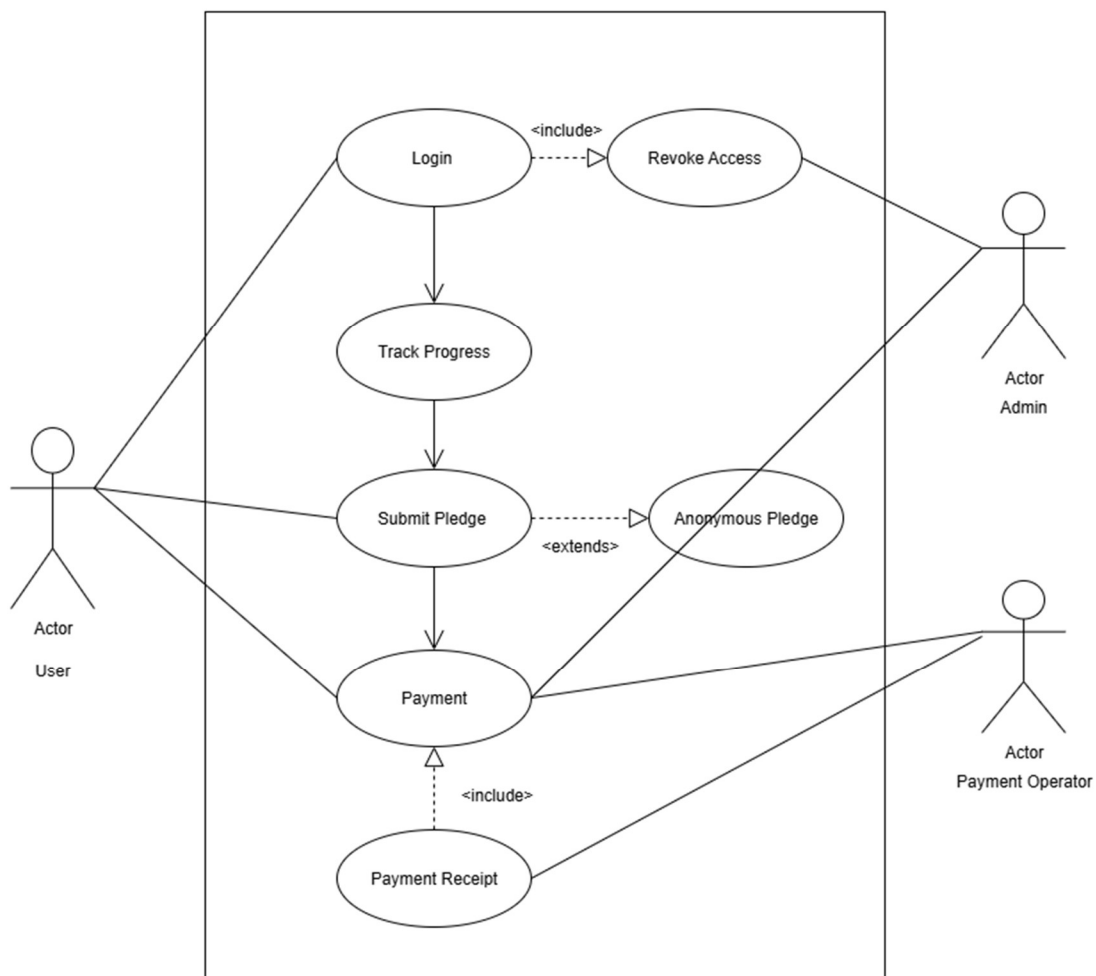
6. Quality attributes & Acceptance tests

Exit criteria for acceptance: All high-priority functional and non-functional requirements implemented and verified, no unresolved critical bugs, and RTM shows all test cases pass.

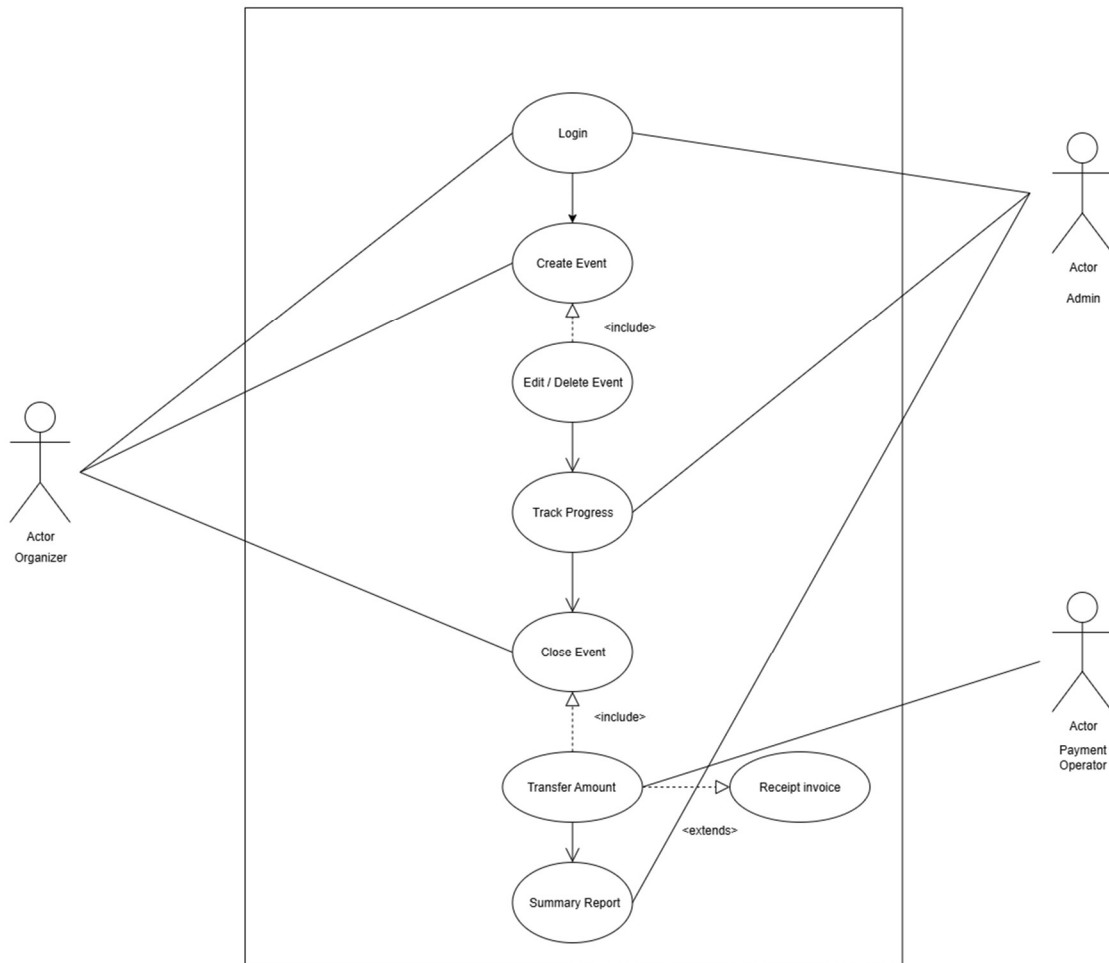
Acceptance test suites: Event Management, Donor Pledge Submission, Progress Tracking, Reports & Analytics, Security, Backup & Recovery.

7. System models and diagrams

7.1 UML Use-Case diagram (User)



7.2 UML Use-Case diagram (Organizer)



8. Requirements Traceability Matrix (RTM)

8.1 Functional Requirements Traceability Matrix

1. Event Management

Req ID	Requirement	Module	Test Case	Status
CFT-F-001	Allow organizers to create	Event Management	TC-Event-01	N

	fundraising events			
CFT-F-002	Allow organizers to edit or delete events before closing	Event Management	TC-Event-02	N
CFT-F-003	Allow organizers to close events once complete or end date reached	Event Management	TC-Event-03	N

2. Donor Pledges

Req ID	Requirement	Module	Test Case	Status
CTF-F-004	Donors can submit pledge via secure online form	Donor Pledge	TC-Pledge-01	N
CTF-F-005	Validate pledge data (amount > 0, donor details required)	Donor Pledge	TC-Pledge-02	N
CTF-F-006	Update event progress in real time after pledge	Progress Tracking	TC-Pledge-03	N

3. Progress Tracking

Req ID	Requirement	Module	Test Case	Status
CFT-F-007	Aggregate total pledges for each event	Progress Tracking	TC-Track-01	N
CFT-F-008	Display progress bar	Progress Tracking	TC-Track-02	N

	showing % completion against target			
CFT-F-009	Generate visual reports (pie, bar graphs) for donation trends	Progress Tracking	TC-Track-03	N

4. User Role and Access

Req ID	Requirement	Module	Test Case	Status
CTF-F-010	Organizer login to access dashboard	User Access	TC-Org-Login-010	N
CTF-F-011	Users can change/reset password securely	User Access	TC-Password-011	N
CTF-F-012	Admin can revoke user access immediately	User Access	TC-Revoked-012	N

5. Reports and Analysis

Req ID	Requirement	Module	Test Case	Status
CFT-F-013	Export donor pledge lists	Reports & Analytics	TC-LIST - 013	N
CFT-F-014	Generate Top Donors report	Report Module	TC-Report-014	N
CFT-F-015	Event summary report	Report Module	TC-SUM-015	N

8.2 Non-Functional Requirements Traceability Matrix

Req ID	Requirement	Module	Test Case	Status
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CFT-NF-001	System availability 99.5% excluding maintenance	System Reliability	TC-Avail-01	N
CFT-NF-002	Pledge submission response time $\leq 2s$	Performance	TC-Perf-01	N
CFT-NF-003	Handle concurrent access from 10,000 donors	Scalability/Performance	TC-Person - 01	N
CFT-NF-004	Daily automated backup of system data	Backup & Recovery	TC-BKP-01	N
CFT -NF- 004	Adaptive Response	User Feasibility	TC-UX-01	N